# Bálint Gyevnár

Carnegie Mellon University
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## Employment

*09/2025 – present* **Carnegie Mellon University**
Postdoctoral Research Associate with Nihar Shah and Atoosa Kasirzadeh
Current projects: assessment of author understanding at grecaptcha.com; investigating failure modes of AI scientist systems, such as indirect data poisoning and backdooring;

## Education

*09/2021 – 05/2026* **University of Edinburgh**
Ph.D. in Natural Language Processing
Supervisors: Stefano Albrecht, Shay Cohen, Chris Lucas
Thesis: Action Explanation of Multi-Agent Systems via Counterfactual Reasoning [link]
Examiners: Ann Nowé, Bonan Zhao

*09/2016 – 05/2021* **University of Edinburgh**
M.Inf in Informatics
Supervisor: Maria Wolters
Thesis: Comparison of Account Survival on Twitter During the First Wave of COVID-19

*08/2018 – 06/2019* **Nanyang Technological University**
Exchange Year in Computer Science

## Past Employment

*07/2023 – 12/2024* **University of Edinburgh**, Edinburgh Futures Institute
Research Assistant with Shannon Vallor, Atoosa Kasirzadeh

*09/2020 – 09/2025* **University of Edinburgh**, School of Informatics
Teaching Assistant (see *Teaching*)

*05/2020 – 10/2020* **FiveAI**, Edinburgh
Research Intern with Stefano Albrecht

## Awards

*2024* **Colours Award for Outstanding Volunteering Contribution to Sports**
Edinburgh University Sports Union

*2023* **AI100 Early Career Essay Competition, Featured Essay**
One Hundred Year Study on Artificial Intelligence, Stanford University

*2022* **Trustworthy Autonomous Systems Early Career Researcher Award**
UK Research & Innovation £4,000

*2021* **Shape the Future of ITS Competition Award**
IEEE Intelligent Transportation Systems Society $1,000

## Preprints

Google Scholar: https://scholar.google.com/citations?user=fLyES3oAAAAJ
Semantic Scholar: https://www.semanticscholar.org/author/Balint-Gyevnar/2007920131
* denotes equal contribution.

[P4] **Bálint Gyevnár** and Simon DeDeo. Autonomous AI Agents Systematically Diverge From Human Preferences in Proof Formalization. *Submitted to Nat. Mach. Intell., 2026.* [link]

[P3] Justin Payan*, **Bálint Gyevnár***, Atoosa Kasirzadeh, and Nihar B. Shah. greCAPTCHA: Assessing Understanding as Evidence of Research Authorship Under Generative AI. *Under review at CHI, 2026.* [link]

[P2] **Bálint Gyevnár**, Atoosa Kasirzadeh, and Nihar B. Shah. Distributed Denial of Science: How Indirect Data Poisoning of AI Systems Can Industrialize Scientific Fraud. *Submitted to PNAS, 2026.* [link]

[P1] **Bálint Gyevnár** and Atoosa Kasirzadeh. Bridging the Gap in the Responsible AI Divides. *Under revision at Patterns, 2026.* [link]

## Peer-Reviewed Publications

### Journal Publications

[J4] **Bálint Gyevnár*** and Atoosa Kasirzadeh*. AI Safety for Everyone. *Nature Machine Intelligence, 2025.* [link]

[J3] Anton Kuznietsov*, **Bálint Gyevnár***, Cheng Wang, Steven Peters, and Stefano. V. Albrecht. Explainable AI for Safe and Trustworthy Autonomous Driving: A Systematic Review. *IEEE Transactions on Intelligent Transportation Systems, 2024.* [link]

[J2] **Bálint Gyevnár***, Gautier Dagan*, Coleman Haley*, Shangmin Guo*, and Frank Mollica. Communicative Efficiency or Iconic Learning: Do Communicative and Acquisition Pressures Interact to Shape Colour-Naming Systems? *Entropy, 2022.* [link]

[J1] Ibrahim H. Ahmed, Cillian Brewitt, Ignacio Carlucho, Filippos Christianos, Mhairi Dunion, Elliot Fosong, Samuel Garcin, Shangmin Guo, **Bálint Gyevnár**, Trevor McInroe, Georgios Papoudakis, Arrasy Rahman, Lukas Schäfer, Massimiliano Tamborski, Giuseppe Vecchio, Cheng Wang, and Stefano V. Albrecht. Deep Reinforcement Learning for Multi-Agent Interaction. *AI Communications, 2022.* [link]

### Conference Publications

[C6] **Bálint Gyevnár**, Christopher G. Lucas, Stefano V. Albrecht, and Shay B. Cohen. Integrating Counterfactual Simulations with Language Models for Explaining Multi-Agent Behaviour. *AAMAS 2026.* [link]

[C5] **Bálint Gyevnár**, Stephanie Droop, Tadeg Quillien, Shay B. Cohen, Neil R. Bramley, Christopher G. Lucas, and Stefano V. Albrecht. People Attribute Purpose to Autonomous Vehicles When Explaining Their Behavior: Insights From Cognitive Science for Explainable AI. *CHI 2025.* [link]

[C4] **Bálint Gyevnár**, Cheng Wang, Christopher G. Lucas, Shay B. Cohen, and Stefano V. Albrecht. Causal Explanations for Sequential Decision-Making in Multi-Agent Systems. *AAMAS 2024.* [link]

[C3] **Bálint Gyevnár**, Nick Ferguson, and Burkhard Schafer. Bridging the Transparency Gap: What Can Explainable AI Learn From the AI Act? *ECAI 2023.* [link]

[C2] Cillian Brewitt, **Bálint Gyevnár**, Samuel Garcin, and Stefano V. Albrecht. GRIT: Fast, Interpretable, and Verifiable Goal Recognition with Learned Decision Trees for Autonomous Driving. *IROS 2021.* [link]

[C1] Stefano V. Albrecht, Cillian Brewitt, John Wilhelm, **Bálint Gyevnár**, Francisco Eiras, Mihai Dobre, and Subramanian Ramamoorthy. Interpretable Goal-based Prediction and Planning for Autonomous Driving. *ICRA 2021.* [link]

### Workshop Publications

[W2] **Bálint Gyevnár**[*] and Mark Towers[*]. Objective Metrics for Human-Subjects Evaluation in Explainable Reinforcement Learning. *RLDM 2025.* [link]

[W1] **Bálint Gyevnár**, Massimiliano Tamborski, Cheng Wang, Christopher G. Lucas, Shay B. Cohen, and Stefano V. Albrecht. A Human-Centric Method for Generating Causal Explanations in Natural Language for Autonomous Vehicle Motion Planning. *IJCAI 2022 Workshop on Artificial Intelligence for Autonomous Driving.* **Best paper runner-up.** [link]

## Teaching

### Teaching Assistant (University of Edinburgh)

*Spring 2024* EFIE11173: Evaluating Sustainable Lands and Cities
EFIE11179: Data, Mobility, Infrastructure

*Spring 2024* INFR11010: Reinforcement Learning

*Autumn 2023* INFR11194: Doing Research in Natural Language Processing

*Spring 2022* INFR11202: Machine Learning Theory

*Autumn 2020* INFR08027: Informatics 2C - Introduction to Computer Systems
INFR10069: Introductory Applied Machine Learning.

### Mentoring

*09/2026–present* **Kerissa Duliga**; Master's student of Computer Science

*09/2026–present* **Yiming Shao**; Undergraduate student of Computer Science

*10/2026–present* **Brandon Zhou**; Master's student of Philosophy

## Service

*10/2025* **Workshop Organiser**
Evaluating Explainable AI and Complex Decision-Making, ECAI 2025

**Reviewer / PC Member**
*Journals*: Nature Communications (2025); Artificial Intelligence Review (2025); International Journal of Human-Computer Interaction (2026); IEEE Transactions on Intelligent Transportation Systems (2025); Journal of Autonomous Agents and Multi-Agent Systems (2026)
*Conferences*: AAAI (2026); AIES (2025); CHI (2025, 2026); ECAI (2023); FAccT (2026, 2027); ICRA (2025); IASEAI (2026, 2027); IROS (2025); NeurIPS (2025); XAI (2026)
*Workshops*: Academic Mindtrek (2026), EMNLP BlackboxNLP (2025), NeurIPS AI4Metascience (2026)

## Media Coverage

*07/2026* Nature News. Coverage of *Distributed Denial of Science*. [link]
*Doctored data sets could trick AI agents*

*07/2026* El País. Coverage of *Distributed Denial of Science*. [link]
*El problema de los clips ya está aquí: las IA pueden destruir el mundo sin querer*

*09/2026* Times Higher Education. Frances Jones. Coverage of *greCAPTCHA*. [link]
*Researchers submitting AI-assisted papers they "don't understand"*

*07/2025* Qubit. Coverage of *AI Safety for Everyone*. [link]
*Igazságkereső AI helyett MuskGPT lett a milliárdos chatbotjából, a Grokból*

## Invited Talks

*06/2026* **Columbia University**. Machine Learning Summer School. (Hosted by Gary Kazantsev)

*03/2026* **IVADO**. Workshop on Social Reasoning and the Ecology of Thought. (Hosted by James Evans)

*06/2025* **University of California, Berkeley**. 9th Workshop of the Center for Human-Compatible AI

*07/2024* **Charles University and Czech Technical University**. Machine Learning and Modelling Seminar. (Hosted by Martin Holeňa)